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The distribution of match activities relative to the maximal intensities in elite soccer players: implications for practice

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ABSTRACT

To determine the distribution of match-activities relative to maximum-intensities during official match in elite soccer players. One hundred and forty-eight Italian Serie A soccer players were monitored during 46 official matches (680 individual-samples). Total distance (TD), high-speed running (HSR), very high-speed running (VHSR), sprint, acceleration and deceleration were calculated. Maximum-intensities (1-min$_{peak}$) were used as the reference value to determine the distribution of relative intensity across the whole-match demands (90-min$_{avg}$). Time and distance higher than 90-min$_{avg}$ (>90-min$_{avg}$) were also calculated. The relative (m·min^{-1}) 90-min$_{avg}$ *vs*1-min$_{peak}$ was 59.6(4.4)% for TD, 26.2(4.4)% for HSR, 16.0 (3.5)% for VHSR, 9.3(2.3)% for sprint, 19.2(4.6)% for acceleration and 15.4(5.2)% for deceleration. Total distance covered >90-min$_{avg}$ was ~61.4(5.0)% for TD, ~68.6(1.9)% for HSR, ~80.2(1.3)% for VHSR, ~95.7(0.4)% for sprint, ~75.5(1.3)% for acceleration and ~64.0 (2.6)% for deceleration. With the exception of *small* [ES: 0.50 (0.26 to 0.73)] difference for acceleration, the relative distance >90-min$_{avg}$ was *largely* to *very largely* (ES: 1.64 to 7.78) higher (P< 0.05) than the 90-min$_{avg}$ for each metric. While no between-position difference (P> 0.05) was found for total minutes >90-min$_{avg}$, between-position differences (P< 0.05) for the total distance >90-min$_{avg}$ were retrieved across each metric. The distribution of the activities relative to maximal intensities could assist coaches for soccer training prescriptions.

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Team sports; training load; performance; coaching; match analysis; peak demands; worst case scenario

Introduction

In soccer, understanding the activities of match-play is crucial for soccer-specific training prescriptions (Johnston et al., 2020; Martin-Garcia et al., 2018b; Riboli et al., 2020, 2021b). The quantification of the physical demands using different tracking technologies (e.g., global positioning system, semi-automatic video-analysis, etc.) is used currently to determine the total distance (TD), the distance covered at different running speed (Di Salvo et al., 2007) and acceleration/deceleration during both training and match (Martin-Garcia et al., 2018a). In practice, for preparing performance across consecutive matches, the activities recorded during the competitive games are used to plan the training workload and as a reference for soccer-specific drills (e.g., small-sided games), technical-tactical

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drills and/or individual positional exercises (Lacome et al., 2018; Martin-Garcia et al., 2018b; Riboli et al., 2020).

Although the most common time-motion analysis has been focused on fixed periods of 90-min (Di Salvo et al., 2007), 15-min (Bradley et al., 2010) and 5-min (Bradley & Noakes, 2013)match demands, a rolling method was suggested to avoid underestimation in the most demanding passages of match play (MDP) (Martin-Garcia et al., 2018a; Riboli et al., 2021b; Varley et al., 2012). It was suggested that during the training process the relative whole-match running distances (90-min_{avg}) fail to fully account for the MDP (Riboli et al., 2021b; Varley et al., 2012) and the attention has been focused on MDP over different lengths of time (e.g., 1- to 10-min) (Casamichana et al., 2019; Cunningham et al., 2018; Ferraday et al., 2020). For a mere example, the distance covered >24 $km{\cdot}h^{-1}$ was ~5.8, ~7.3, ~11.9, ~23.3 and ~42.0 $m{\cdot}min^{-1}$ for 90-min, 10-min, 5-min, 2-min and 1-min peak (1-min_{peak}) demands, respectively (Riboli et al., 2021b). Therefore, 90-min_{avg} may be responsible for underpreparing players for the MDP (Cunningham et al., 2018; Delaney et al., 2017; Riboli et al., 2021b).

The 1-min_{peak} during official elite soccer matches were investigated in Serie A (Riboli et al., 2021b), French Ligue 1 (Lacome et al., 2018), English Championship (Ferraday et al., 2020) and reserve squad Spanish La Liga (Casamichana et al., 2019; Martin-Garcia et al., 2018a). Moreover, the occurrence of 1-min_{peak} over the match (i.e., 1st half *vs* 2nd half) (Casamichana et al., 2019), the effect of formation (Riboli et al., 2021b), ball-in-play time (Riboli et al., 2021b; Wass et al., 2020) and ball or no ball possession phases (Riboli et al., 2021b) were investigated for helping practitioners to contextualize 1-min_{peak} across different soccer-specific contexts. It was showed that the locomotor requirements during the 1-min_{peak} was about fourfold higher than 90-min_{avg}, especially for the distance >24 $km{\cdot}h^{-1}$ (+624% than 90-min_{avg}) (Riboli et al., 2021b). Hence, recreating the MDP across different time-durations and 1-min_{peak} during the training sessions allows the players to be conditioned for the maximal demands of competition, which is not taken into account when analysing the 90-min_{avg} only (Oliva-Lozano et al., 2020; Riboli et al., 2021b; Varley et al., 2012; Wass et al., 2020).

Notwithstanding the maximum running performances theoretically occur only once or few times during the game (Varley et al., 2012). Consequently, conditioning for the MDP should be only a part of the overall periodized training programmes. A recent study in Australian football and rugby league showed that the greatest volume of activity was at ~60% of the 1-min_{peak} demands (Johnston et al., 2020). The authors suggested that a detailed description of the match intensity distribution rather than only 1-min_{peak} may be useful for practitioners faced with the day-by-day prescriptions of skill-based exercises (Johnston et al., 2020). However, despite the distance covered in a different range of speed thresholds may be used to provide a more comprehensive assessment of the intermittent running demands typically occurring in team sports (Riboli et al., 2020; Young et al., 2018), only total distance and acceleration/deceleration distance were investigated (Johnston et al., 2020). In elite soccer, although 90-min_{avg} and 1-min_{peak} guide the prescription of the training intensity, the distribution and the training volume of match activities relative to the maximum intensities is still unknown. For example, despite the 90-min_{avg}, 1-min_{peak} or other time-durations (e.g., 2-, 3-, 5-min, etc.) can be utilized to modulate locomotor demands during soccer-specific drills, positional-drills and/or running-based exercises (Lacome et al., 2018; Riboli et al., 2020, 2021b), the quantification of

the volume, sets and/or duration that should be utilized during practice to recreate or to overload such sport-specific match demands are still unknown. Therefore, the current study aimed to determine the distribution of different locomotor activities with regard to 1-min$_{peak}$ demand during official matches in elite soccer players. Additionally, to determine the time spent at different intensities relative to the 1-min$_{peak}$ demand. Lastly, to describe the time spent and the distance covered at each locomotor demand higher than the 90-min$_{avg}$ (i.e., from 90-min$_{avg}$ to 1-min$_{Peak}$) across different metrics.

Materials and methods

Participants

One-hundred and forty-eight (*n = 148*) Italian Serie A soccer players were monitored during matches across the 2019–2020 season. Within each match, the players were classified according to playing position, resulting in the following number of data sets per position: central defenders (*n = 36*), wide defenders (*n = 16*), central midfielders (*n = 35*), wide midfielders (*n = 19*), wide forwards (*n = 17)*, and forwards (*n = 25*). Particularly, a given player was analysed in the position he actually played in that game. Goalkeepers were not included in the analysis. A total of *n* = 680 individual observations were collected. The number of individual matches varied from home players [*n* = 20.6(9.5), *range*: 38–3)] to opposite players [*n* = 1.4(0.7), *range*: 4–1)]. The Ethics Committee of the University of Milan (protocol #102/14) approved the study. It was performed in accordance with the principles of the Declaration of Helsinki (1975).

Experimental design

Data were collected during 46 official home matches in the same stadium from both home and opposition players at each match. The same stadium was used to avoid possible bias due to the placement of different tracking systems in different stadia, possibly influencing the data. A semi-automatic tracking system (Stats Perform, Chicago, Illinois, USA) was used to quantify players' running performance. The validity and reliability of this system have been previously established (Buchheit et al., 2014).

Procedures

Following the completion of each match (~90 min), each file was trimmed so that only data recorded when the player was on the field for at least 85 min were included for further analysis. Data were exported into a customized Microsoft Excel spreadsheet (Microsoft, Redmond, USA). A customized spreadsheet was used to allow analysis of relative distance covered (m·min^{-1}) in the following categories: total distance (TD), high-speed running distance (HSR, 15 to 20 km·h^{-1}), very high-speed running distance (VHSR, 20 to 24 km·h^{-1}), sprint distance (sprint, >24 km·h^{-1}), and distance with variations in running speed >3 m·s^{2} (acceleration and deceleration) (Riboli et al., 2020). To assist in the development of velocity-based movement indicators, rolling moving average was utilized to calculate the most demanding one-minute period (1-min$_{Peak}$) for each player across each match with the maximum value collected (Casamichana et al., 2019; Cunningham

et al., 2018; Riboli et al., 2021b). To compare with the traditional metrics analysis, the distances over the whole match demand (90-min$_{avg}$) was recorded and inserted into the data analysis. The 1-min$_{peak}$ were then used as the reference value to determine the distribution of relative intensity across the whole match for all other rolling 1-min periods. The time and distance higher than 90-min$_{avg}$ (>90-min$_{avg}$) were calculated as the minutes or distance covered at intensity higher than the percentage of 1-min$_{peak}$ corresponding to the average 90-min$_{avg}$.

Statistical analysis

SPSS (version 26, Chicago, IL, USA) was used to perform the statistical analysis. A linear mixed model analysis was used to compare the effects of position x distribution of the match activities on the dependent parameter (Abt & Lovell, 2009; Riboli et al., 2021b). The model used for each dependent parameter was with position and distribution of match activities as independent fixed factors and random intercepts on the individual players. A log-likelihood ratio test was used to assess the goodness of fit of the models. Bonferroni's correction was used for multiple comparison analysis. Cohen's *d* effect size (ES) with 95% confidence intervals (CI) was used to describe the magnitude of the pairwise differences and interpreted as follows: <0.20: *trivial*; 0.20–0.59: *small*; 0.60–1.19: *moderate*; 1.20–1.99: *large*; ≥2.00: *very large* (Hopkins et al., 2009). Statistical significance was set at $\alpha < 0.05$. Unless otherwise stated, all values are presented as mean (SD) as reported using descriptive statistics.

Results

As showed in Figure 1, the 90-min$_{avg}$ *vs* 1-min$_{peak}$ was on average at 59.6(4.4)% for TD, 26.2(4.4)% for HSR, 16.0(3.5)% for VHSR, 9.3(2.3)% for sprint, 19.2(4.6)% for acceleration

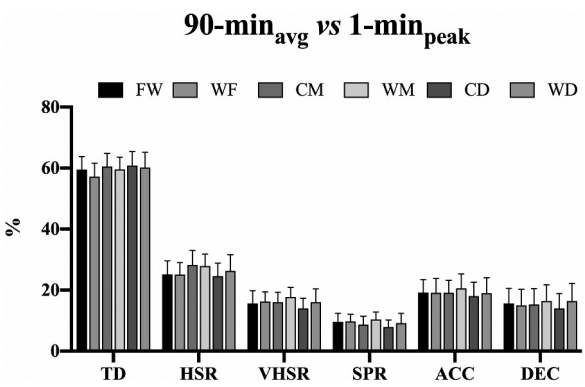

Figure 1. The match activities of the average 90-min official match demands (90-min$_{avg}$) are shown as the percentage of the 1-min peak (1-min$_{peak}$) for each position across each metric. FW: forwards; WF: wide forwards; CM: central midfielders; WM: wide midfielders; CD: central defenders; WD: wide defenders. The relative distance covered at total distance (TD), high-speed running (HSR), very-high speed running (VHSR), sprint (SPR), acceleration (ACC) and deceleration (DEC) is shown.

and 15.4(5.2)% for deceleration. No between-position differences ($P > 0.05$) were found across each metric.

The distribution of the time spent at different percentages of 1-min$_{peak}$ for TD, HSR, VHSR, sprint, acceleration and deceleration across each position is reported in Figure 2. Main effects for distribution of match activities were found for each dependent parameter ($P < 0.001$). No main effect for factor position was observed ($P < 0.05$). Some *trivial* to *small* (ES: 0.11 to 0.49) between-position differences were found and reported in Figure 2. The average distribution of the metre covered at different percentages of 1-min$_{peak}$ for TD, HSR, VHSR, sprint, acceleration and deceleration is reported in Table 1. Main effects for distribution of locomotor demands were found for each dependent parameter ($P < 0.001$).

Table 2 summarizes the time spent and the total distance covered >90-min$_{avg}$ for TD, HSR, VHSR, sprint, acceleration and deceleration; the percentage of the total distance covered during the whole match at >90-min$_{avg}$ is also reported. The

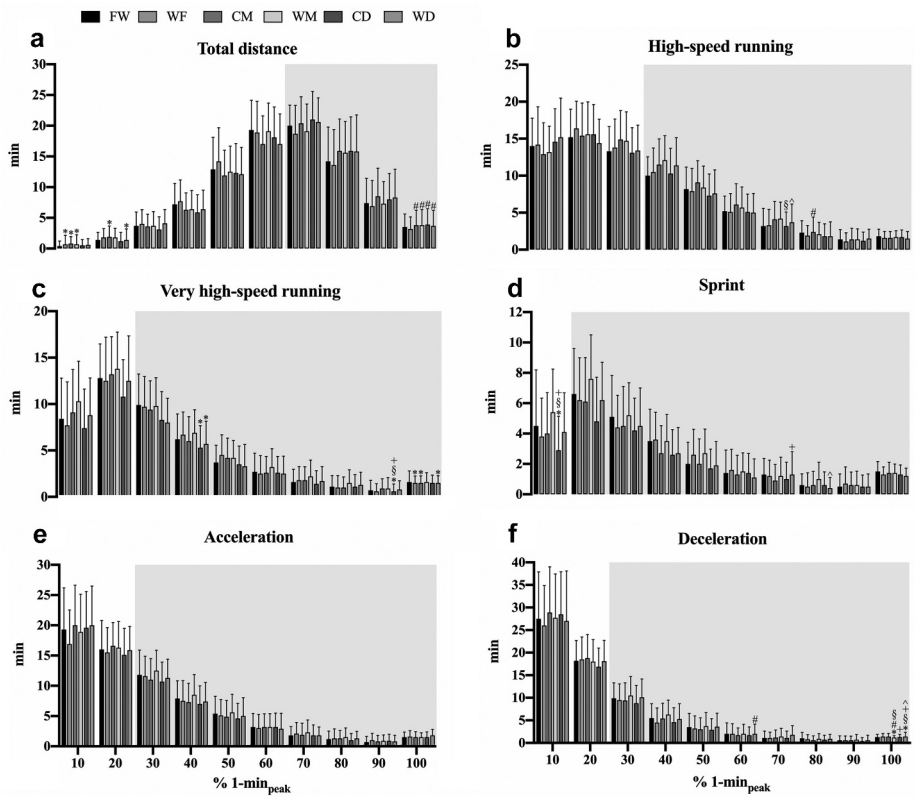

Figure 2. The time spent at different percentages (from 0–10% to 90–100%) of the peak demands recorded in 1 min (% 1-min$_{peak}$) is shown for each position for each metric. The grey area highlights the time spent at match activities higher than the average whole-match demands. Total distance: Panel A; high-speed running: Panel B; very-high speed running: Panel C; sprint: Panel D; acceleration: Panel E; deceleration: Panel F. FW: forwards; WF: wide forwards; CM: central midfielders; WM: wide midfielders; CD: central defenders; WD: wide defenders. $^{*}P < 0.05$ *vs* FW; $^{\#}P < 0.05$ *vs* WF; $^{\S}P < 0.05$ *vs* CM; $^{+}P < 0.05$ *vs* WM; $^{\wedge}P < 0.05$ *vs* CD.

Table 1. The distance covered at different percentage relative to the 1-min$_{peak}$ for each locomotor metrics is shown.

%1-min$_{peak}$	TD	HSR	VHSR	Sprint	Acceleration	Deceleration
0–10%	9.7 (2.4)	116.1 (4.9)	30.6 (6.0)	17.3 (4.7)	20.8 (2.1)	49.5 (3.3)
10–20%	57.4 (11.1)	282.1 (27.2)	90.6 (12.9)	53.0 (11.3)	39.1 (3.3)	67.1 (5.3)
20–30%	204.8 (24.2)	396.6 (57.0)	100.9 (11.7)	57.2 (10.1)	41.9 (4.2)	52.9 (6.7)
30–40%	493.6 (50.0)	431.2 (66.9)	91.4 (14.4)	52.0 (11.7)	37.4 (5.0)	35.1 (6.7)
40–50%	1188.2 (99.1)	390.0 (59.2)	71.9 (11.1)	45.2 (12.0)	31.7 (3.9)	26.2 (3.8)
50–60%	2069.9 (105.5)	308.6 (48.9)	55.6 (9.2)	35.0 (5.6)	22.2 (1.9)	17.2 (1.5)
60–70%	2580.2 (146.4)	236.5 (54.9)	44.4 (9.8)	33.6 (7.1)	16.3 (2.5)	13.6 (2.7)
70–80%	2211.6 (249.0)	155.0 (28.8)	33.9 (8.0)	21.0 (5.7)	11.9 (2.0)	9.7 (1.9)
80–90%	1254.1 (157.7)	112.1 (22.8)	23.1 (5.4)	19.8 (4.9)	8.2 (0.8)	7.5 (1.7)
90–100%	679.8 (86.3)	167.1 (15.4)	57.6 (6.0)	57.3 (5.8)	19.6 (1.3)	27.1 (1.1)

TD: total distance; HSR: high-speed running; VHSR: very high-speed running. Bold text highlights the distance covered at percentages higher than the average percentage corresponding to the 90-min match demands.

Table 2. The time spent and the distance covered at each locomotor activity >90-min$_{avg}$ is reported as mean (SD). The percentage the distance covered during the whole match >90-min$_{avg}$ is also reported.

	TD	HSR	VHSR	Sprint	Acceleration	Deceleration
Minutes	46(2)	33(2)	27(2)	21(2)	33(2)	24(2)
Meters	6725.8(620.7)	1800.3(269.3)	478.8(71.3)	374.1(81.1)	189.2(14.0)	189.4(18.1)
>90-min$_{avg}$ vs whole match	61.4(5.0)%	68.6(1.9)%	80.2(1.3)%	95.7(0.4)%	75.5(1.3)%	64.0(2.6)%

TD: total distance; HSR: high-speed running; VHSR: very high-speed running.

differences ($P < 0.001$) between the relative distance covered at >90-min$_{avg}$ and the relative 90-min$_{avg}$ were *small* (ES: 0.50; CI: 0.26 to 0.73) for acceleration [5.7(0.3) m·min^{-1} *vs* 5.3(0.8) m·min^{-1}], *large* (ES: 1.91; CI: 1.64 to 2.19) for deceleration [7.9(0.6) m·min^{-1} *vs* 6.1(1.1) m·min^{-1}] and *very large* (ES: 2.40 to 7.78) for TD [146.1(7.2) m·min^{-1} *vs* 126.5(7.2) m·min^{-1}], HSR [54.5(6.5) m·min^{-1} *vs* 15.4(3.1) m·min^{-1}], VHSR [17.7(1.8) - m·min^{-1} *vs* 8.2(1.5) m·min^{-1}] and sprint [17.8(2.7) m·min^{-1} *vs* 5.7(1.3) m·min^{-1}].

Figure 3 summarizes the time spent and the total distance covered >90-min$_{avg}$ for TD, HSR, VHSR, sprint, acceleration and deceleration across each position. No between-position difference ($P > 0.05$) was found across each metric for the time spent >90-min$_{avg}$. Conversely, the between-position differences ($P < 0.05$) for the total distance covered at >90-min$_{avg}$ were *trivial* to *moderate* (ES: 0.12 to 1.16) for TD, *trivial* to *large* for HSR (ES:0.06 to 1.21) and VHSR (ES: 0.06 to 1.97), *small* to *large* (ES: 0.21 to 2.16) for sprint, *trivial* to *small* (ES: 0.02 to 0.45) for acceleration and *small* to *large* (ES: 0.35 to 1.37) for deceleration.

Discussion

The current study was designed to determine the distribution of the match activities during official matches with regard to 1-min$_{peak}$ in elite soccer players. Firstly, the locomotor activities calculated at 90-min$_{avg}$ were much lower than 1-min$_{peak}$, especially for the high-intensity activities. Secondly, most locomotor activities occur at an intensity >90-min$_{avg}$. Thirdly, with the exception of the *small* difference for acceleration, the relative distance >90-min$_{avg}$ was *largely* to *very largely* higher than the 90-min$_{avg}$ for each metric. Additionally, the time spent at >90-min$_{avg}$ showed no between-position differences, while between-position differences in the total distance covered at >90-

min_{avg} appeared across each metric. The present findings provide new insights about the distribution of the intensity of the locomotor activities across the matches.

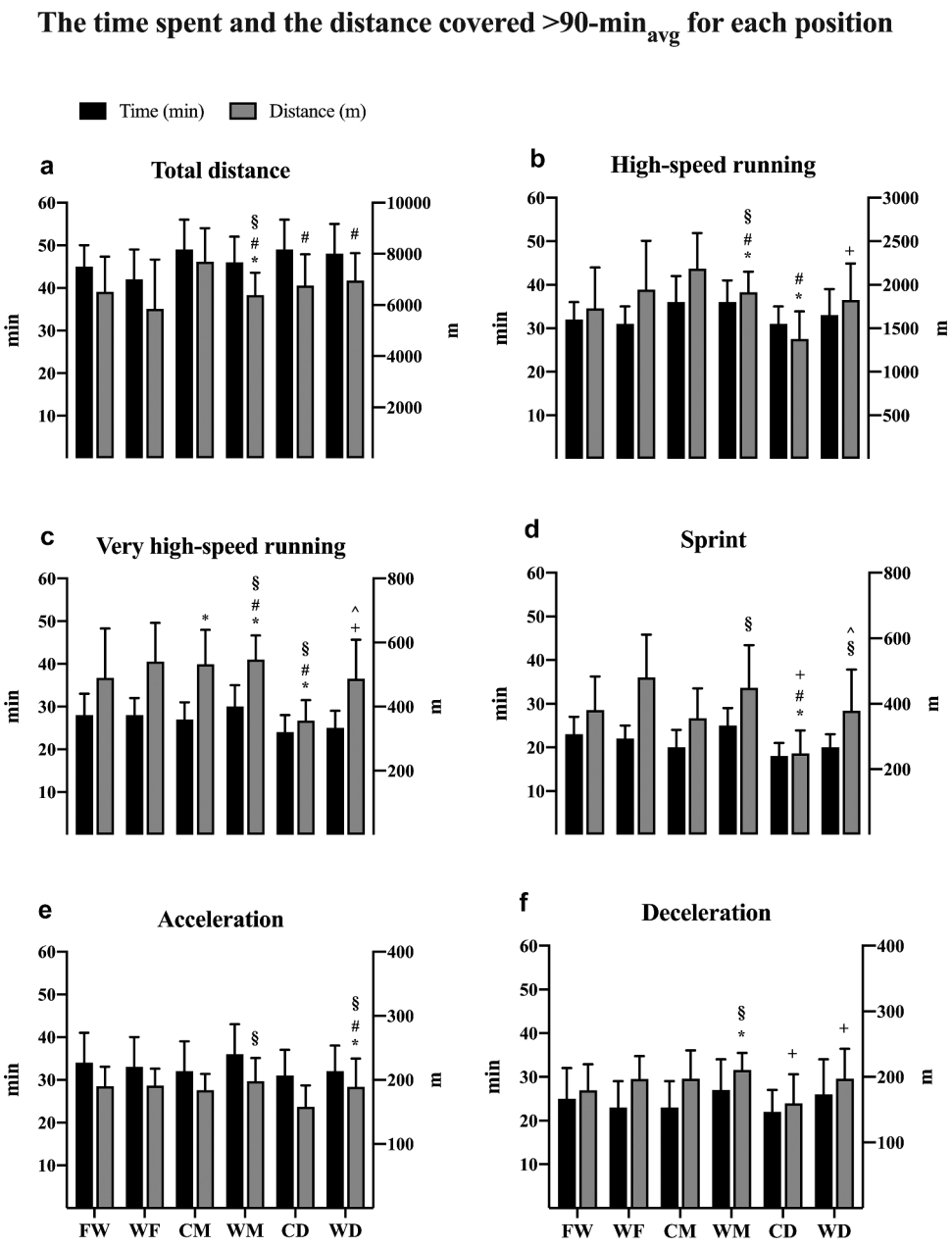

Figure 3. The time spent (min) and distance covered (m) at match activities higher than the average whole-match demand (90-min_{avg}) are shown for each position. Total distance: Panel A; high-speed running: Panel B; very-high speed running: Panel C; sprint: Panel D; acceleration: Panel E; deceleration: Panel F. FW: forwards; WF: wide forwards; CM: central midfielders; WM: wide midfielders; CD: central defenders; WD: wide defenders. $^{*}P < 0.05$ *vs* FW; $^{\#}P < 0.05$ *vs* WF; $^{\S}P < 0.05$ *vs* CM; $^{+}P < 0.05$ *vs* WM; $^{\wedge}P < 0.05$ *vs* CD.

The present findings remarkably highlighted the gap between 90-min$_{avg}$ and 1-min$_{peak}$ especially for the high-intensity activities in elite soccer players, as previously shown (Riboli et al., 2021b). At this regard, the relative 90-min$_{avg}$ fails to fully account for the peak demands that occur during official matches in different team sports, such as rugby (Delaney et al., 2017), Gaelic football (Malone et al., 2017), hurling (Young et al., 2018), Australia football (Johnston et al., 2020) and soccer (Ferraday et al., 2020) and it may be responsible for underpreparing players for the maximum intensities of the competitions (Lacome et al., 2018; Riboli et al., 2021b; Varley et al., 2012). Moreover, it was shown that soccer-specific contextual factors such as the ball-in-play time, ball/no ball possession phases and/or different formations seems to affect the 1-min$_{peak}$ (Riboli et al., 2021b), suggesting that practitioners should vary the training stimuli accordingly. The gap between 90-min$_{avg}$ and 1-min$_{peak}$ should be considered with the intention of replicating or overloading match demands using soccer-specific drills.

The distribution of the time spent at different percentages of 1-min$_{peak}$ for TD, HSR, VHSR, sprint, acceleration and deceleration across each position in elite soccer were described for the first time. Therefore, a comparison with previous findings is challenging. A recent study in Australian football and rugby league (Johnston et al., 2020) reported that the greatest volume of the match activity for TD was similar to the present findings (i.e., ~60% than 1-min$_{peak}$), while further direct comparisons cannot be made because of the different metrics used. As such, most match activities seem to occur at an intensity greater than what usually determined when considering the whole-match demands. This may depend on the inclusion of time-windows during which players walk, stand or perform low-intensity activities, thus lowering the mean match demands. For example, sprint distance showed ~96% of the whole distance covered at intensity >90-min$_{avg}$, highlighting such a limitation when only mean values are used. Additionally, with the exception of the *small* difference for acceleration, *large* to *very large* differences in relative 90-min$_{avg}$ *vs* >90-min$_{avg}$ were found for each metric. No between-position differences for the time spent at match activities >90-min$_{avg}$ were found, while between-position differences in the total distance covered >90-min$_{avg}$ suggest that some players should be conditioned differently. Despite the same time spent >90-min$_{avg}$, central-defenders and wide-defenders showed a lower distance covered than other positions, especially for HSR, VHSR and sprint. In practice, positional-specific references should be utilized as a minimal reference to overload such metrics with regards to the maximal intensities of the competition.

The match demands across different time-windows were previously proposed as minimal references for soccer-specific exercises lasting different time-durations (Lacome et al., 2018; Martin-Garcia et al., 2018a; Riboli et al., 2021b). The present findings suggest that practitioners may consider 1-min$_{peak}$, the time spent at different percentages of 1-min$_{peak}$, the time spent >90-min$_{avg}$ and the total/relative distance >90-min$_{avg}$ for soccer-specific training prescriptions in elite players. It should be however in mind that the maximal running performances (1-min$_{peak}$) theoretically occur only once or few times during the game (Varley et al., 2012). Therefore, this worst-case scenario should be only a part of the overall periodized training prescriptions. Notwithstanding knowledge of the time spent at different percentages of 1-min$_{peak}$, may help practitioners to adequately periodize training

programmes that expose athletes to official-match demands with regards to 90-min_{avg}, >90-min_{avg} and maximum match activities. In the practice of the soccer-specific drills such as small- or large-sided games (Riboli et al., 2020) and/or individual position-specific soccer drills (Bradley et al., 2019), different percentages of 1-min_{peak} may be used to vary the stimuli and prepare the players for the actual match demands.

Limitations

It should be acknowledged that the maximal individual capacity in different metrics (e.g., VHSR, sprint, acceleration, deceleration, etc.) can exceed the maximal positional match requirements. For example, locomotor load >90-min_{avg} or 1-min_{peak} for a given player could be lower than his VHSR, sprint, acceleration or deceleration maximal capacity. For training prescription purposes, considering locomotor load >90-min_{avg} or 1-min_{peak} only may lead to lower training stimuli that could not effectively condition the players (Riboli et al., 2020, 2021b). Therefore, soccer-specific exercises (e.g., small- or large-side games) (Riboli et al., 2020), position-specific drills (Ade et al., 2020) and/or individualized running-based exercises (Riboli et al., 2017) with the aims to recreate or overload locomotor load from >90-min_{avg} to 1-min_{peak} should be coupled with soccer-specific, positional-specific or individual running-based exercises near to the maximal individual aerobic (Riboli et al., 2021a), anaerobic (Bangsbo, 2015)and neuromuscular (Zouhal et al., 2019) capacity for maximizing the performance development in elite soccer players. Additionally, no internal training load was assessed, as this is difficult to assess during official matches. Moreover, the number of high-intensity activities such as sprint, acceleration or decelerations and directional changes was not examined, and we acknowledge that this would add more practical information to the practitioners, so that future studies are needed. Lastly, the game location, the ranking of opposition, the congested periods and the seasonal periods need to be further investigated.

Conclusions

The present study examined the distribution of the activities relative to the maximal intensities (1-min_{peak}) during official matches in elite soccer players. The locomotor activities calculated at 90-min_{avg} were much lower than 1-min_{peak}, especially for the high-intensity activities (corresponding to ~19% to ~9% of the 1-min_{peak}). Additionally, especially for the high-intensity activities, most of the time is spent above the average match demands. Position differences in the total distance covered at intensity >90-min_{avg} are visible and should be considered for training prescription.

Practical applications

- Reduce the training/match gap using soccer-specific exercises (e.g., small- or large-side games), position-specific soccer drills and/or running-based exercises with regard to the actual maximal intensities of the competition.
- The training practice should take into account all the intermediate intensities between the average and peak match demands.

- The time spent and the distance covered >90-min$_{avg}$ can be used as a reference for training intensity in relationship with its duration.
- The distribution of the time spent at different percentages of 1-min$_{peak}$ can be used as reference for the durations, repetitions and/or sets that should be utilized during practice to recreate or to overload such sport-specific metrics. As a mere example, ~6 min at intensity higher than 80% of the 1-min$_{peak}$ for the distance >20 km·h^{-1} could be used as a minimal reference to plan at least 6 repetitions of 1-min using soccer-specific positional drills during practice.
- Forwards and midfielders should be conditioned differently to fulfil the peak demands.

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Disclosure statement

The authors have no conflicts of interests to declare.

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